

Discrete Structure

CSIT — 2nd Semester — 2082 — Answer Sheet
Subject Code: CSC165

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. State inclusion ■ exclusion principle. For which number x is true that $[x] = [x]$? Let f and g be functions from $\{1, 2, 3, 4\}$ to $\{a, b, c, d\}$, and from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$, respectively with $f(1)=d$, $f(2)=c$, $f(3)=a$, and $f(4)=b$, and $g(a)=2$, $g(b)=1$, $g(c)=3$ and $g(d)=2$. a. If f and g one ■ to ■ one? b. If f and g on ■ to? c. Does either f or g have an inverse? If so find this inverse.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How many ways are there to assign 24 students to five faculty advisors? Find the solution of the recurrence relation $a_n = 3a_{n-1} + 3a_{n-2} + a_{n-3}$ if $a_0=2$, $a_1 = 2$ and $a_2 = 4$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define an Euler circuit and Euler path in an undirected graph. Compute the maximal flow form following network flow.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Explain how the pigeonhole principle can be used to show that among any 11 integers, at least two must have the same last digit. Find a minimum spanning tree from following graph where the degree of each vertex in the spanning tree does not exceed 2.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Find the solution to the system $x = 1 \pmod{4}$, $x = 2 \pmod{5}$ and $x = 3 \pmod{7}$ using Chinese Remainder theorem.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Suppose that on an island there are three types of people, knights, knaves, and normals. Knights always tell the truth, knaves always lie, and normals sometimes lie and sometimes tell the truth. Detectives questioned three inhabitants of the island—Amy, Brenda, and Claire—as part of the investigation of a crime. The detectives knew that one of the three committed the crime, but not which one. They also knew that the criminal was a knight, and that the other two were not. Additionally, the detectives recorded these statements: Amy: "I am innocent." Brenda: "What Amy says is true." Claire: "Brenda is not a normal." After analyzing their information, the detectives positively identified the guilty party. Who was it?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Use mathematical induction to prove that $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Suppose that R is the relation on the set of strings of English letters such that aRb if and only if $L(a) = L(b)$, where $L(x)$ is the length of the string x . Is R an equivalence relation?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is the negation of "This is a boring course"? Give a direct proof for the statement: "If n is even, then $n + 4$ is even."

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define well ordering property. Find the multiplicative inverse of 7 modulo 19 using Extended Euclidean Algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What are the necessary conditions for graph to be isomorphic? List and illustrate with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Solve the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.